Gaze Reliability for Mixed Reality: Domain-Conditioned Guarantees for Attention-Aware XR Systems

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Abstract

Eye tracking is increasingly integrated into mixed reality (MR) systems to support attention-aware applications, where gaze serves as a direct proxy for visual attention and is highly correlated with task performance. However, in real-world deployments, network disruptions such as rendering instability and dynamic latency perturbations actively alter and degrade user gaze behavior. This degradation manifests as elevated gaze divergence and reduced dwell times, ultimately compromising task effectiveness. To address this, we present a unified framework designed to prevent gaze degradation by maintaining MR systems within verifiable safe operating ranges. Our approach introduces two core technical contributions. First, we develop a time-aware gaze forecasting model that explicitly handles irregular and asynchronous MR telemetry by parameterizing temporal gaps between observations. Second, we introduce a domain-conditioned statistical certification framework that provides rigorous guarantees for decision-relevant attention properties, such as timely orientation to objects of interest (OOIs) and sustained fixation. Rather than merely evaluating predictive regression accuracy, we compute exact binomial confidence intervals across observable scene and system statistics to identify operating domains that preserve baseline-consistent gaze behavior. We evaluate our framework on an MR tactical search-and-rescue testbed under induced latency perturbations. Empirical results demonstrate that our system yields actionable, high-confidence guarantees, e.g., certifying at least an 0.82 probability of sustained OOI attention under bounded system stress at 90% confidence. By restricting system operation to these certified domains, our framework serves as a deployable reliability layer that dynamically mitigates service disruptions to preserve user attention and mission integrity.

Index Terms: Mixed reality, gaze prediction, attention-aware systems, robustness, statistical certification, cognitive attacks.

1 Introduction

Eye tracking is rapidly becoming a standard sensing capability in mixed and extended reality (MR/XR), enabling gaze-based interaction, attentive user interfaces, and rendering optimizations that explicitly depend on where a user is looking [1, 2, 3]. In these systems, gaze is not merely a logging signal: it is frequently incorporated into the runtime decision loop (e.g., selection, guidance, content adaptation, and gaze-contingent rendering), so deviations in gaze behavior can directly change what the system does.

In many visually guided tasks, gaze serves as an observable proxy for overt visual attention, and eye-movement measures provide actionable indicators of information sampling and task engagement. Classic and modern evidence supports the link between attention and eye movements (e.g., attentional mechanisms in saccadic behavior and task-driven gaze allocation) [4, 5]. In applied settings, eye-based sampling measures have been shown to correlate with task performance, motivating the use of gaze-derived metrics (e.g., time-to-orientation and dwell/fixation behavior) as performance-relevant signals [6, 7]. As a result, maintaining "safe" (i.e., non-degraded, baseline-consistent) gaze behavior is a practical goal for sustaining attention-aware MR functionality.

However, MR systems rarely operate under ideal conditions. Real deployments experience service disruptions and runtime perturbations—such as motion-to-photon latency, latency jitter, and frame-time instability—arising from resource contention, network variability, or adversarial manipulation of system performance [8, 9, 10, 11]. These perturbations can desynchronize or sparsify telemetry streams, and eye-tracking pipelines themselves are subject to latency and sampling constraints [2, 12]. Because gaze is used inside the control loop, the key reliability question becomes not only whether gaze can be estimated or predicted on average, but whether the system can preserve baseline attention-relevant gaze behaviors when operating conditions degrade.

Prior work has explored gaze tracking, interaction, and gaze prediction in VR/XR, including benchmark efforts that evaluate short-horizon gaze prediction [13] and systems that anticipate gaze for latency-sensitive gaze-contingent techniques [14, 15]. Related streams also model future focus of attention to enable predictive rendering and reduce perceived latency in edge-assisted VR pipelines [16]. Yet, these approaches are typically assessed via aggregate prediction error or end-to-end quality metrics, and they rarely provide explicit, runtime-checkable guarantees that mission-level gaze properties (e.g., timely orientation to objects of interest) remain stable under service disruptions.

This paper addresses the following research question:

Under service disruptions, what operating regions of measurable MR system parameters preserve stable and baseline-relative gaze behavior?

To answer this question, we present a unified framework that connects MR operating conditions to expected gaze behavior and enables mitigation by constraining system operation within a safe operating range.

First, we develop a time-aware gaze forecasting model that produces gaze predictions with respect to the combination of MR system operation parameters and disruption conditions. The model explicitly accounts for irregular and asynchronous MR telemetry by incorporating elapsed-time gaps between observations into its state update, following the general principle that time gaps and missingness should be treated as first-class signals rather than resampled away [17, 18, 19, 20].

Second, we introduce a domain-conditioned certification procedure that provides statistical guarantees for attention-relevant gaze properties. Using measurable scene and system statistics, we identify operating domains ("safe operating ranges") in which we can certify, with high confidence, that baseline-consistent gaze behavior is maintained. Our statistical certification relies on established binomial confidence bounds as conservative guarantees of property satisfaction [21, 22, 23].

Finally, under disruptions, we use the predictive model to reason mitigation strategies by restricting MR system operation within a constrained parameter range (our safe operating range) to ensure that the same gaze behaviors are maintained. We evaluate this framework in an MR search-and-rescue testbed under induced latency perturbations, demonstrating that (i) gaze metrics are positively correlated with task performance and (ii) safe-range operation preserves baseline-consistent gaze behavior despite external

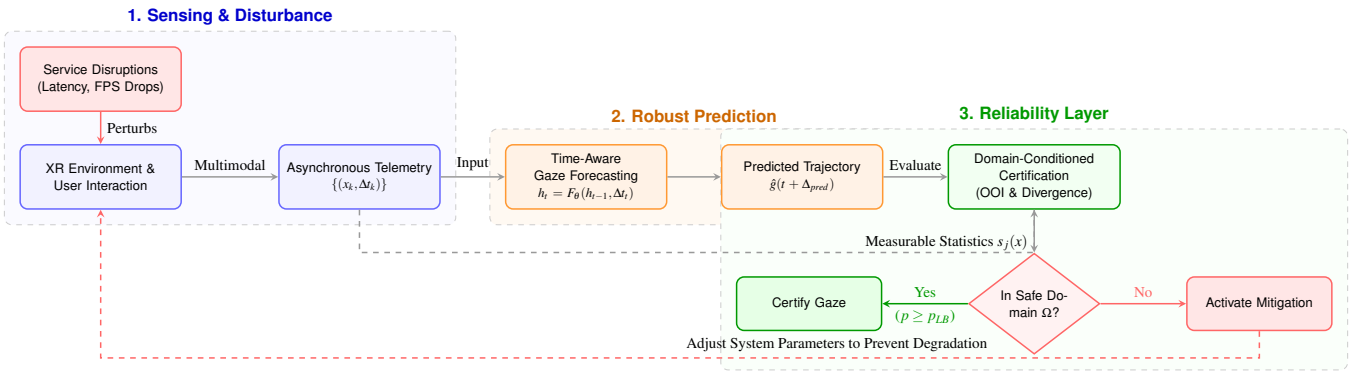

Figure 1: Overview of the proposed domain-conditioned gaze reliability framework. (1) The XR system experiences service disruptions, generating irregular and asynchronous telemetry. (2) A time-aware sequence model forecasts gaze trajectories by explicitly parameterizing temporal gaps Δt_t. (3) The runtime reliability layer evaluates these predictions against measurable scene statistics to determine if the system remains within a certified safe operating domain Ω. If out-of-domain, the system dynamically triggers mitigation strategies to prevent gaze degradation and sustain task performance.

perturbations.

We evaluate the framework on an MR tactical search-and-rescue testbed under induced latency perturbations. Our empirical analysis confirms that gaze metrics positively correlate with task performance, justifying the need to prevent degradation. Furthermore, our results demonstrate that when the MR system operates within the identified safe operating ranges, user gaze behavior remains consistent with nominal, undisrupted conditions, enabling a deployable XR reliability layer.

The main contributions of this work are summarized as follows:

- **Empirical analysis of gaze behavior under system perturbations.** We analyze how gaze metrics such as orientation latency and gaze dwell time change under varying MR system conditions and show that these metrics correlate with task performance in a search-and-rescue scenario.

- **Robust time-aware gaze forecasting for irregular MR telemetry.** We develop a gaze prediction model that explicitly incorporates temporal gaps and multimodal system telemetry, enabling robust gaze forecasting under irregular sampling and runtime perturbations.

- **Operating-domain reasoning for maintaining gaze behavior.** Rather than evaluating gaze prediction solely through regression accuracy, we define decision-relevant properties such as timely orientation to OOIs. We then introduce a framework that identifies safe operating ranges of system parameters under which attention-related gaze behaviors remain consistent with baseline operation.

- **Experimental validation in a mixed-reality mission scenario.** We evaluate the framework on a mixed-reality search-and-rescue testbed with induced perturbations, showing that maintaining operation within identified domains preserves baseline-consistent gaze behavior.

Together, these results suggest that monitoring and maintaining stable gaze behavior provides a practical mechanism for sustaining attention-aware interaction in mixed reality systems operating under real-world perturbations. The rest of the paper is organized as follows. Section 2 reviews related literature. Section 3 formalizes the telemetry model and defines the service disruption perturbation process. Section 4 details the time-aware gaze forecasting architecture. Section 5 introduces the domain-conditioned certification framework. Section 6 describes the MR testbed and the experimental evaluation protocols, while Section 7 presents the empirical results. Finally, Section 8 and Section 9 discuss and conclude the paper, repectively.

2 Related Work

2.1 Eye Tracking in XR and Gaze as an Attention-Relevant Signal

Eye tracking enables interaction, adaptive content presentation, and user-state inference in XR, and has been surveyed extensively in recent literature [1, 2, 3]. These surveys document a broad design space where gaze is used for direct interaction (e.g., selection and assistance) and for system-level optimizations (e.g., gaze-contingent rendering), implying that gaze is often part of the runtime decision loop.

From the vision and human-factors perspective, eye movements provide observable evidence about overt visual attention and task-driven information sampling. Prior work connects attention mechanisms to saccadic behavior and emphasizes task-dependent gaze allocation in natural behavior [4, 5]. Moreover, eye-based sampling measures have been shown to correlate with task performance in dynamic monitoring tasks, motivating gaze-derived metrics as performance-relevant signals [6]. In HCI and ergonomics, gaze measures have also been used to characterize user strategies and interface effectiveness [7]. Our work builds on this foundation but asks a different question: not how to use gaze under nominal conditions, but how to maintain baseline-consistent gaze behavior when MR operating conditions are perturbed.

2.2 Gaze Prediction and Gaze-Contingent Techniques in VR/XR

Gaze prediction has been studied for latency-sensitive XR techniques, including benchmarks that evaluate multi-frame-ahead gaze-vector prediction in head-mounted settings [13]. Anticipatory gaze models have also been used to compensate for delays in gaze-contingent rendering, including approaches that predict saccade landing positions to mitigate perceptible lag [15]. More recent systems predict future focus of attention to enable predictive (pre-)rendering, particularly in edge-assisted VR pipelines where latency and bandwidth constraints motivate anticipatory allocation of resources [16]. Related work also studies real-time gaze prediction from head pose and gaze histories for short prediction horizons in VR architectures [14].

Our paper differs in problem formulation and evaluation target. Rather than optimizing prediction accuracy alone, we model gaze as a function of both MR system operation parameters and disruption conditions, and we use gaze predictions to derive runtime-checkable safe operating ranges that preserve mission-level gaze behaviors.

2.3 Modeling Irregular Telemetry and Providing Guarantees

A central challenge in deployed MR is that telemetry is heterogeneous and can become irregular under perturbations (e.g., frame-time instability, pipeline delays). Latency and timing instability are known to affect VR/AR usability, safety, and user experience [8, 10, 11, 9]. Additionally, sampling characteristics can materially affect eye-tracking measures, making irregularity itself a modeling concern [12].

To handle irregular temporal dynamics, we draw on time-aware sequence modeling that explicitly encodes elapsed time and missingness patterns, including Time-Aware LSTM [17], GRU-D [18], and Phased LSTM [19]. Continuous-time latent dynamics (e.g., latent ODE formulations) are also a principled approach to arbitrary observation gaps [20]. These works motivate our design choice to incorporate time gaps and stale context as first-class inputs.

On the assurance side, our statistical certification uses established binomial confidence bounds [21, 22, 23] to provide conservative guarantees for attention-relevant properties over operating domains. This differs from formal verification approaches for neural networks (including for recurrent models) that aim to prove properties of model outputs under bounded perturbations [24, 25]. Our focus is instead on identifying runtime-checkable operating domains over measurable MR statistics, enabling mitigation by constraining system operation to preserve baseline gaze behavior under service disruptions.

3 System Model and Problem Formulation

In mixed reality systems equipped with eye tracking, gaze signals provide a continuous observation of where a user directs visual attention. XR telemetry streams are often irregular and asynchronous. System perturbations such as frame-rate fluctuations may influence both sensor measurements and model predictions. In this section, we formalize the gaze forecasting problem, define mission-level attention properties that capture attention-relevant behaviors, and introduce operating domains that enable runtime reliability certification.

3.1 XR Telemetry and Irregular Sampling

Let t denote an anchor time, and let W be an observation-window duration. Because MR sensing and rendering are irregular, we observe a variable-length sequence of samples within $[t-W,t]$:

$$\mathscr{H}(t) = \{(x_k, \Delta t_k)\}_{k=1}^{T(t)}, \quad \Delta t_k = t_k - t_{k-1},$$

where $x_k \in \mathbb{R}^d$ is the multimodal telemetry vector of dimension d at timestamp t_k and $T(t)$ is the number of samples observed in $[t-W,t]$. The multimodal vector x_k encapsulates continuous eye and head tracking kinematics, scene statistics, and asynchronous discrete events.

The key systems feature is that Δt_k is not constant. It varies with rendering load, scheduling jitter, communication delay, and system perturbations. The observation model is therefore inherently irregular.

The forecasting task is to predict normalized screen gaze coordinates $\hat{g}(t+\Delta_{\text{pred}}) \in [0,1]^2$ at a future time $t+\Delta_{\text{pred}}$:

$$\hat{g}(t+\Delta_{\text{pred}}) = f_\theta\Big(\{(x_k, \Delta t_k)\}_{k=1}^{T(t)}\Big),$$

where f_θ is a parameterized forecasting model. The prediction horizon Δ_{pred} reflects the latency between sensing and system response in XR applications.

3.2 Service Disruptions and Perturbation Model

Real-world XR deployments are subject to service disruptions such as rendering instability and network latency. We model these disruptions as a perturbation process π that corrupts the nominal history $\mathscr{H}(t)$ into a degraded observation $\mathscr{H}^\pi(t)$.

Rather than merely adding white noise, π structurally alters the telemetry stream in two specific ways. First, it inflates the temporal gaps Δt_k through latency injection and dropped frames. Second, it alters the visual scene statistics embedded within x_k, such as temporal smoothness and optical flow, due to the degraded rendering pipeline.

When downstream XR systems rely on f_θ to infer visual attention, processing $\mathscr{H}^\pi(t)$ instead of $\mathscr{H}(t)$ causes the predicted gaze trajectory to deviate from the user's true intent. To address this challenge, our methodology is twofold. In Section 4, we design a time-aware forecasting model that maintains stable predictions under irregular sampling. In Section 5, we introduce a domain-conditioned certification framework that mathematically guarantees the reliability of gaze-driven decisions even when the system is under perturbation.

4 Time-Aware Gaze Forecasting Model

In practice, XR systems produce asynchronous observations from multiple sources, including head motion, scene dynamics, and system-level statistics. These signals arrive at different sampling rates and exhibit variable time gaps due to scheduling delays, pipeline fluctuations, or service disruptions. A forecasting model that assumes uniform sampling may therefore fail to capture the temporal dynamics of gaze behavior in real deployments. To address this issue, we adopt a time-aware modeling approach that explicitly incorporates temporal intervals and multimodal telemetry into the prediction process, preserving temporal fidelity under variable rendering loads.

4.1 Multimodal Input Representation

At each observation time t, the XR system records a telemetry vector that aggregates contextual information relevant to gaze behavior. Our architecture organizes inputs into two distinct modalities to handle their inherently different sampling behaviors.

First, the continuous numeric stream $x_t^{\text{num}} \in \mathbb{R}^{d_{\text{num}}}$ contains dense measurements. This includes 3D head kinematics, observable scene features such as optical flow and temporal smoothness, and system performance metrics. Second, the system records sparse asynchronous streams for discrete events and object detections. Let $\mathscr{E}$ denote the history of discrete events and $\mathscr{O}$ denote the history of object detections. Because XR telemetry streams are asynchronous, consecutive dense observations are separated by variable time intervals $\Delta t_t = t - t_{\text{prev}}$. The combined tuple of numeric data, sparse contexts, and the temporal gap forms the complete multimodal input representation at time t.

4.2 Asynchronous Sparse Context Encoding

Discrete events and object detections occur irregularly and are often absent during a given frame update. We employ a Last-Observation-Carried-Forward (LOCF) approach augmented with exponential temporal decay to encode these sparse streams.

Suppose the most recent event e^* occurred at timestamp t^*_{evt}. The staleness of this observation at current time t is defined as $s_{\text{evt}} = t - t^*_{\text{evt}}$. We compute a decay factor $\gamma_{\text{evt}} = \exp(-\beta_{\text{evt}} s_{\text{evt}})$, where β_{evt} is a learned decay rate. The event representation $\bar{e}_t$ is then constructed by interpolating between the learned embedding of e^* and a null token embedding based on this decay factor.

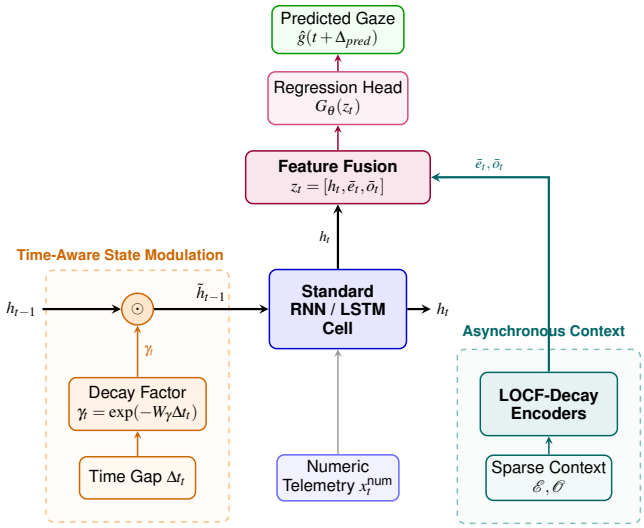

Figure 2: Architecture of the Time-Aware Gaze Forecasting Model. The model handles irregular XR telemetry through two parallel mechanisms. First, the temporal gap Δt_t explicitly modulates the continuous hidden state via a learned decay factor γ_t, preventing stale kinematics from distorting the RNN. Second, sparse and asynchronous streams (events $\mathscr{E}$, objects $\mathscr{O}$) are processed via LOCF-Decay encoders. These modalities are fused into a joint representation z_t to produce a robust gaze prediction even under severe rendering disruptions.

This mechanism ensures that the influence of a discrete XR event naturally fades over time. Object detections are processed using an identical decayed LOCF mechanism to yield an object representation $\bar{o}_t$. These decayed representations preserve critical contextual information without requiring artificial sequence padding.

4.3 Time-Aware Recurrent Modeling

To capture gaze dynamics from the continuous numeric stream under irregular sampling, the forecasting model explicitly incorporates the temporal gap Δt_t into the recurrent state update. Intuitively, the influence of past observations should decay as the time interval between samples increases. Closely spaced observations conversely imply stronger temporal continuity.

Let h_t denote the hidden state of the sequence model at time t. We apply a learned temporal decay to the previous hidden state before incorporating new telemetry [18]. The time-aware decay factor γ_t is defined as

$$\gamma_t = \exp(-W_\gamma \Delta t_t),$$

where W_γ is a learned parameter vector. The decayed hidden state

$$\tilde{h}_{t-1} = \gamma_t \odot h_{t-1}$$

is then supplied to the standard recurrent cell alongside the current numeric input x_t^{num}:

$$h_t = \text{RNN}_\theta(\tilde{h}_{t-1}, x_t^{\text{num}}).$$

This formulation enables the model to attenuate stale information during large temporal gaps while preserving continuous dynamics during dense sampling. Such time-aware recurrent architectures have proven effective for irregular time series in healthcare and event-based vision tasks [17, 26]. In the XR setting, this explicitly protects the hidden state from distortion caused by fluctuating frame rates and latency perturbations.

4.4 Gaze Forecasting and Optimization

Given the temporal state h_t and the sparse context representations $\bar{e}_t$ and $\bar{o}_t$, the model predicts the future gaze location using a fused regression head. We concatenate these representations into a unified feature vector $z_t = [h_t, \bar{e}_t, \bar{o}_t]$. The regression head G_θ maps this joint representation to a normalized gaze coordinate:

$$\hat{g}(t+\Delta_{\text{pred}}) = G_\theta(z_t),$$

where $\hat{g} \in [0,1]^2$. The prediction horizon Δ_{pred} corresponds to the time delay between sensing and system response in XR applications, which typically lies in the range of tens to hundreds of milliseconds.

During training, eye-tracking sensors provide ground-truth gaze measurements $g(t) \in [0,1]^2$. The model parameters θ are optimized by minimizing the Euclidean discrepancy between the predicted and actual gaze coordinates using a mean squared error loss:

$$\mathscr{L} = \|\hat{g}(t+\Delta_{\text{pred}}) - g(t+\Delta_{\text{pred}})\|_2^2.$$

By explicitly incorporating temporal gaps and multimodal context, the model learns the dependencies between human gaze behavior and dynamic XR system states. This robust formulation is strictly necessary for the subsequent certification framework. It ensures baseline predictions remain reliable even when the rendering pipeline degrades, allowing the system to verify attention-relevant behaviors under measurable operating conditions.

5 Domain-Conditioned Certification Framework

The time-aware forecasting model in Section 4 provides predictions of future gaze behavior under observed telemetry conditions. However, prediction accuracy alone does not directly indicate whether the resulting attention inference remains reliable for downstream XR decisions. In practice, system perturbations such as rendering instability may influence both the quality of telemetry and the stability of model predictions. To bridge prediction and system safety, we introduce a domain-conditioned certification framework. The central idea of this framework is to identify measurable operating domains within which specific mission-level attention properties hold with high statistical confidence.

Because the defining variables of these domains are observable scene statistics and system parameters, XR systems can evaluate domain membership online. This allows the system to determine whether gaze-based performance is normal or disturbed at runtime.

5.1 Mission-Level Attention Properties

While gaze forecasting models are often evaluated using generic regression error metrics, XR systems ultimately rely on gaze predictions to infer user attention. We therefore define decision-relevant properties that mathematically describe how gaze interacts with objects of interest (OOIs) and how trajectory predictions respond to service disruptions.

5.1.1 OOI Orientation and Sustained Fixation

OOI Orientation Let $R \subset [0,1]^2$ denote the spatial region corresponding to an object of interest (OOI) on the display. Suppose the OOI appears at time t_0. We say that the orientation property holds if the predicted gaze trajectory enters the OOI region within a strict response interval $[t_0, t_0 + T_{\text{resp}}]$:

$$\exists t \in [t_0, t_0 + T_{\text{resp}}] \quad \text{s.t.} \quad g(t) \in R.$$

This property captures whether the gaze prediction indicates timely visual attention toward an important emerging target.

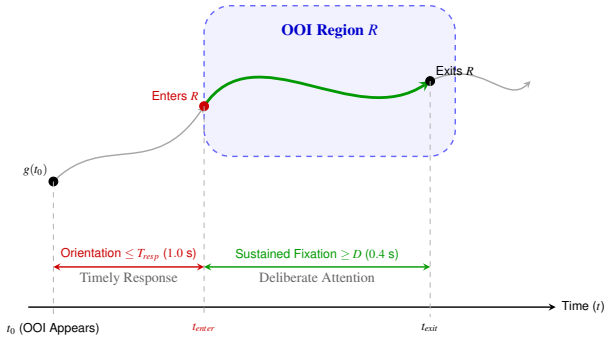

Figure 3: Visualizing Mission-Level Attention Properties. To certify reliable gaze behavior under system disturbances, we define two decision-relevant metrics. First, the predicted gaze trajectory $g(t)$ must orient to the Object of Interest (OOI) region R within a designated response window T_{resp} (e.g., 1.0 s). Second, the gaze must maintain sustained fixation inside R for a minimum dwell duration D (e.g., 0.4 s) to indicate deliberate visual attention rather than a transient glance.

Sustained Fixation Beyond initial orientation, effective situational awareness requires deliberate fixation. Many XR tasks require that the user maintain visual attention on the object long enough for reliable interpretation or interaction. Let $\text{dwell}(g,R)$ denote the cumulative duration for which the gaze remains inside region R. The sustained fixation property holds if

$$\text{dwell}(g,R) \geq D,$$

where D is a minimum dwell-time threshold.

Empirical analysis of our MR dataset reveals that the mean observed dwell duration for initial OOI orientation is 0.22 seconds. In our evaluation, we deliberately enforce a strict 0.4-second sustained focus threshold ($D = 0.4$) alongside a 1.0-second reaction window ($T_{\text{resp}} = 1.0$). By nearly doubling the empirical mean, we ensure that the certified property represents deliberate, mission-relevant visual attention rather than a transient or accidental glance.

5.1.2 Gaze Divergence Under Perturbation

To quantify the sensitivity of gaze forecasting under adversarial conditions such as frame-rate degradation, we consider two gaze trajectories under different operating conditions. Let $\hat{g}_{\text{base}}(t)$ denote the gaze point at time t under nominal system conditions with a stable frame rate. Let $\hat{g}_{\text{perturbed}}(t)$ denote the predicted gaze point at time t under the perturbed telemetry history $\mathscr{H}^{\pi}(t)$. The divergence between the two trajectories at each timestamp t is defined by the Euclidean distance:

$$D(t) = \|\hat{g}_{\text{perturbed}}(t) - \hat{g}_{\text{base}}(t)\|_2. \tag{1}$$

The window-averaged divergence over a specific evaluation window T is computed as:

$$\overline{D}_{[t_0,t_0+T]} = \frac{1}{N} \sum_{t\in[t_0,t_0+T]} D(t). \tag{2}$$

We aim to bound this time-averaged divergence strictly below a threshold τ. This ensures that users' gaze deviations remain insensitive to system perturbations and reliable for downstream reasoning. In our evaluation, we define $\tau = \beta r$, where $r = 0.08$ in normalized coordinates. This radius corresponds to an isotropic pixel margin of approximately 165 pixels on a standard 2304×2064 display, roughly approximating the foveal region of the user's view.

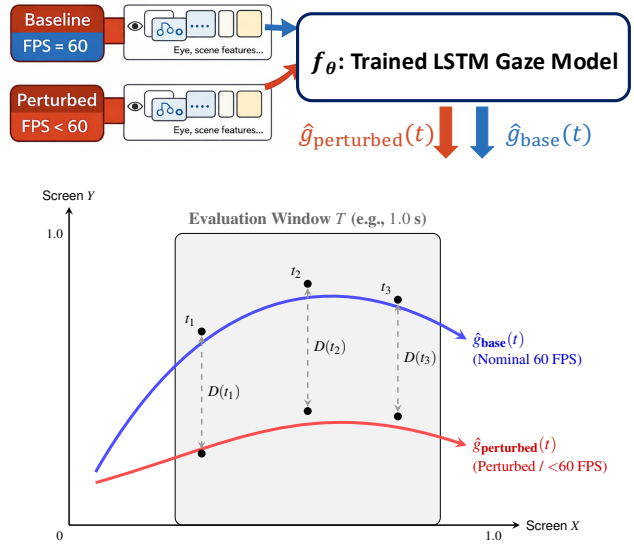

Figure 4: Quantifying Gaze Divergence under Service Disruptions. To evaluate robustness, we compare the gaze trajectory predicted under nominal conditions ($\hat{g}_{\text{base}}$) against the trajectory predicted under perturbed telemetry ($\hat{g}_{\text{perturbed}}$). The instantaneous spatial distance $D(t)$ in (1) is computed at each timestamp. To certify stable gaze behavior, the framework guarantees that the window-averaged divergence $\overline{D}$ in (2) remains bounded below a decision-relevant threshold τ.

5.2 Operating Domain Search and Definition

The reliability of attention inference depends entirely on observable scene statistics and system conditions. We define an operating domain Ω as a bounded subset of the telemetry space characterized by constraints on measurable statistics:

$$\Omega = \{x : L_j \leq s_j(x) \leq U_j,\ j = 1,\ldots,m\}.$$

Here $s_j(x)$ denotes a specific statistic computed from the telemetry window (e.g., optical flow magnitude, brightness flicker, spectral entropy, spatial frequency, or temporal smoothness). The interval $[L_j, U_j]$ specifies the admissible range for that statistic.

To identify optimal bounding intervals $[L_j, U_j]$ that maximize system coverage while preserving safety, the framework executes a data-driven constraint search protocol. The system evaluates candidate domains by ranking them using an approximate Wilson score interval to quickly prune unverified regions. The top candidate domains then proceed to rigorous statistical certification.

5.3 Statistical Certification

For a fixed candidate operating domain Ω, we evaluate whether an attention property holds with high probability. Let n denote the total number of evaluation windows whose telemetry lies inside Ω, and let k denote the number of those windows in which the target attention property is successfully satisfied.

We treat these outcomes as Bernoulli trials with an underlying success probability p. To obtain a mathematically conservative guarantee on p, we compute the exact Clopper-Pearson lower confidence bound [21]:

$$p_{\text{LB}} = \text{CP}_{\text{lower}}(k,n,\alpha),$$

which satisfies the probability inequality:

$$\Pr(p \geq p_{\text{LB}}) \geq 1-\alpha.$$

Throughout our evaluation, we report lower confidence bounds at a confidence level of $1-\alpha = 0.90$. This certification provides a statistically valid guarantee. Whenever the measurable XR system state operates inside the certified domain Ω, the required attention property holds with a probability of at least p_{LB}.

5.4 Runtime Reliability Layer

The certification results directly enable a deployable runtime reliability layer for gaze-aware XR systems. At runtime, the system performs a continuous three-step verification loop:

1. Compute scene statistics and system parameters for the current telemetry window.

2. Determine whether the current state lies safely within a certified operating domain Ω.

3. If the state is in-domain, seamlessly enable gaze-driven adaptation and decision-making. If the state is out-of-domain, immediately activate mitigation strategies such as target enlargement, rendering stabilization, or conservative interaction policies.

This framework is consistent with Fig. 1, which identifies the measurable regions of system parameters required for stable gaze behavior. By mathematically preventing gaze degradation, it serves as a robust foundation upon which future MR mitigation mechanisms can be confidently built.

6 Experimental Setup

We evaluate the proposed framework using a mixed-reality search-and-rescue testbed subjected to induced system perturbations. The evaluation assesses both the predictive accuracy of the time-aware forecasting model and the statistical validity of the domain-conditioned reliability layer.

6.1 TSR Testbed and Dataset

The experimental data originates from a Tactical Search and Rescue (TSR) mixed-reality testbed. We utilize data from 8 user sessions where participants completed ten-minute MR training missions. During these sessions, users experienced dynamic latency conditions representing severe service disruptions.

The system captured synchronized multimodal streams including eye tracking (e.g., position, gaze direction), inertial head tracking, sparse discrete event logs (e.g., tool usage), sparse object detections, and real-time scene video features. The extracted scene statistics include brightness flicker, optical flow, spectral entropy, spatial frequency, and temporal smoothness. Telemetry is irregularly sampled with a nominal rate around 60 Hz. We normalize all gaze coordinates to $[0,1]^2$, which maps to a 2304×2064 display resolution.

6.2 Disruption Protocol

The perturbation protocol systematically varies dynamic latency conditions by dropping the frame rate (FPS) every minute. These controlled disruptions alter rendering continuity and induce irregular telemetry. This provides a rigorous realization of the perturbation model introduced in Section 3.

The maximum available frame rate of 60 FPS defines the nominal baseline. Lower FPS values define increasingly severe perturbation bins. Session boundaries are precisely defined by the timestamps of the first disruption onset and the final mission action.

6.3 Forecasting Evaluation and Baselines

We process the telemetry using time-based windows of duration $W = 5$ s to predict gaze at a horizon of $\Delta_{\mathrm{pred}} = 100$ ms. Training anchors are spaced at least 50 ms apart, while validation and test anchors are spaced 100 ms apart. Label alignment relies on linear interpolation with a maximum tolerance of $\delta_{\max} = 50$ ms.

We conduct a leave-one-user-out (LOUO) cross-validation across the 30 users. Each fold isolates one complete group for testing and uses the remaining groups for training. Validation data is drawn exclusively from temporally separated tail segments of the training groups to prevent temporal data leakage.

Metrics. The primary forecasting metric is the normalized 2D Root Mean Square Error (RMSE):

$$\mathrm{RMSE}_{\mathrm{norm}} = \sqrt{\frac{1}{N}\sum_{n=1}^{N}\left((g_x^{(n)} - \hat{g}_x^{(n)})^2 + (g_y^{(n)} - \hat{g}_y^{(n)})^2\right)}.$$

We also report pixel-space RMSE and the axis-wise coefficient of determination (R^2).

Baselines. To demonstrate the necessity of time-aware modeling under irregular telemetry, we compare our proposed architecture against three baselines. The first is a Standard LSTM that processes sequential inputs but strictly ignores temporal gaps. The second is a Constant-Velocity predictor that linearly extrapolates the most recent gaze trajectory. The third is a static Persistence baseline that simply assumes the future gaze location equals the most recently observed gaze coordinate.

6.4 Mission-Level Certification Protocols

To evaluate human task performance and system reliability, we define specific protocols for the attention properties defined in Section 5.

Behavioral Metrics. We compute First Fixation Time (FFT) and Gaze Dwell Duration (GDD) relative to the appearance of mission-critical targets. Mission effectiveness is defined through rescue latency, measured as the total elapsed time required to successfully complete a rescue interaction.

OOI Property Protocol. For each OOI appearance, successful orientation requires the predicted gaze to enter the OOI-centered region within 1.0 s and remain inside for at least 0.4 s. The dataset contains 417 total object appearances. Empirical analysis shows 67 cases entering the target region within 1.0 s with a mean observed dwell duration of 0.22 s. The chosen 0.4 s dwell threshold is intentionally stricter than the empirical mean. This ensures the certified metric represents deliberate visual attention rather than a transient or accidental glance.

Divergence Protocol. To evaluate robustness against service disruptions, we compare the perturbed predictions against a 60 FPS nominal baseline using the identical telemetry history. The evaluation windows are 1.0 s long with a stride of 0.5 s. Success is defined as maintaining the window-averaged divergence $\bar{D}$ below a margin $\tau = \beta r$. We set $r = 0.08$ in normalized units and evaluate $\beta \in \{0.20, 0.25, 0.30\}$. The reported statistical guarantees rely entirely on this mission-scaled thresholding policy.

7 Results and Evaluation

7.1 Empirical Link Between Gaze Degradation and Task Execution

To justify the need for runtime gaze certification, we first empirically demonstrate how service disruptions alter user attention and how these alterations couple with task performance. We analyze human behavior during the search-and-rescue mission using two established metrics. First Fixation Time (FFT) measures the latency

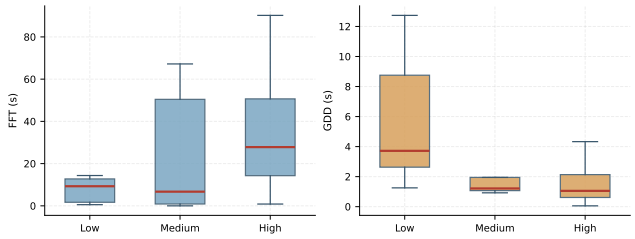

Figure 5: Attack-conditioned human attention distributions. Severe perturbation shifts FFT upward and compresses GDD downward, while also broadening the upper tail of FFT. This indicates that rendering instability degrades both response timeliness and the consistency of sustained attention.

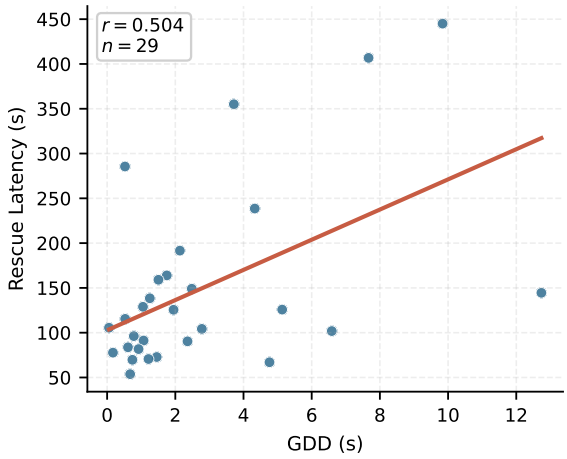

Figure 6: Coupling between rescue latency and cumulative GDD over the hostage-to-rescue interval. The fitted trendline and Pearson statistic show that gaze dwell is not independent of mission execution time. Attention degradation therefore propagates into task-level performance analysis rather than remaining a purely perceptual artifact.

from a hostage's appearance to the first continuous 100-millisecond gaze intersection. Gaze Dwell Duration (GDD) measures the duration of the first continuous fixation exceeding 400 milliseconds, representing deliberate attention allocation.

We examine how these metrics respond to dynamic latency perturbations across three severity bins: low attack (FPS ≥ 40), medium attack (FPS between 20 and 40), and high attack (FPS ≤ 20). The empirical data reveals a severe structural breakdown in visual attention under system stress.

First, service disruptions critically delay target orientation. The mean FFT increases monotonically from 17.20 seconds under low perturbation to 39.82 seconds under high perturbation. Furthermore, the 90th percentile FFT under high perturbation spikes to 80.62 seconds, indicating that severe latency introduces extreme orientation delays rather than a uniform shift in reaction time.

Second, service disruptions shatter sustained attention. The mean GDD collapses from 5.78 seconds under low perturbation to just 1.68 seconds under high perturbation. The upper-tail dwell values similarly contract, confirming that users cannot maintain stable and precise gaze on the object of interest when the rendering pipeline degrades.

Finally, we analyze the relationship between these attention metrics and actual mission effectiveness. We define rescue latency as the total elapsed time from the hostage's first appearance to the successful completion of the rescue event. Pearson correlation analysis reveals a statistically significant coupling between GDD and rescue latency ($r = 0.504$, $p = 0.0053$).

These opposing directional trends where perturbations simultaneously delay initial orientation and destroy fixation stability confirm that network disruptions actively degrade the human cognitive process. Because visual attention is systematically coupled with task execution, protecting gaze behavior from degradation is a fundamental system requirement. This empirical reality necessitates the predictive and certification frameworks evaluated in the following subsections.

7.2 Robust Forecasting Accuracy

To calibrate the difficulty of the 100 ms prediction horizon and demonstrate the necessity of modeling temporal irregularities, we evaluate the proposed architecture against three established baselines across the complete 30-user dataset. Table 1 summarizes the aggregate leave-one-user-out (LOUO) forecasting performance. The Time-Aware LSTM achieves the best performance across all metrics. It significantly reduces the normalized RMSE to 0.059 compared to 0.081 for the Standard LSTM and 0.115 for the Persistence baseline.

The performance gap between the two recurrent variants confirms that explicitly encoding temporal gaps yields substantial and consistent gains over a structurally identical but time-agnostic LSTM. Furthermore, while the Persistence baseline is typically competitive at short horizons due to human gaze inertia, the time-aware model captures deeper predictive structure and kinematic trends. It decisively outperforms simple carry-forward or constant-velocity behaviors, validating the benefit of data-driven multimodal modeling in contested XR environments.

Table 1: Aggregate LOUO cross-validation forecasting performance across 30 users for a 100 ms prediction horizon. Results report the mean and standard deviation (SD) across all user folds.

Forecasting Model	$\text{RMSE}_{\text{norm}}$	RMSE_{px}	R_x^2	R_y^2
Persistence [1]	0.115 ± 0.032	265.0 ± 73.7	-0.452 ± 0.180	-0.380 ± 0.150
Constant-Velocity [1]	0.098 ± 0.028	225.8 ± 64.5	-0.120 ± 0.140	-0.105 ± 0.120
Standard LSTM [27]	0.081 ± 0.021	186.6 ± 48.3	0.150 ± 0.110	0.185 ± 0.130
Time-Aware (Ours)	$\mathbf{0.059 \pm 0.015}$	$\mathbf{128.8 \pm 34.5}$	$\mathbf{0.257 \pm 0.085}$	$\mathbf{0.317 \pm 0.095}$

Beyond aggregate means, a robust XR system must guarantee stable performance across diverse human users. Table 2 details the statistical distribution of the LOUO forecasting errors across the participant population. The narrow gap between the 25th percentile (0.0452) and the 75th percentile (0.0680) demonstrates exceptional cross-user generalization. Although performance naturally varies due to individual physiological differences and varying attack severities, the time-aware architecture tightly bounds the error variance. This tight distribution confirms that the framework provides stable and reliable short-horizon predictions across a broad user population without requiring expensive per-user recalibration.

7.3 Certification of Mission-Level OOI

The first guarantee translates prediction into a mission-level responsiveness claim. Under the specific operating constraints

$$\textit{optical_flow} \in [0.02, 0.07], \qquad \textit{temporal_smoothness} \in [20, 55],$$

the probability that gaze enters the OOI-centered region within 1.0 second and remains there for at least 0.4 seconds is certified to satisfy

$$p \geq 0.82 \qquad \text{at 90\% confidence.}$$

Table 2: Distribution of LOUO forecasting performance across 30 users (100 ms horizon). The narrow interquartile range demonstrates consistent cross-user generalization without per-user calibration.

Statistic	$\text{RMSE}_{\text{norm}}$	RMSE_{px}	R_x^2	R_y^2
Best User (Min Error)	0.0381	87.7	0.580	0.510
25th Percentile	0.0452	104.1	0.350	0.380
Median (50th)	0.0550	126.7	0.280	0.330
75th Percentile	0.0680	156.6	0.190	0.250
Worst User (Max Error)	0.0815	187.7	-0.050	-0.020
Overall Mean ± SD	0.0589 ± 0.012	128.8 ± 27.6	0.257 ± 0.11	0.317 ± 0.14

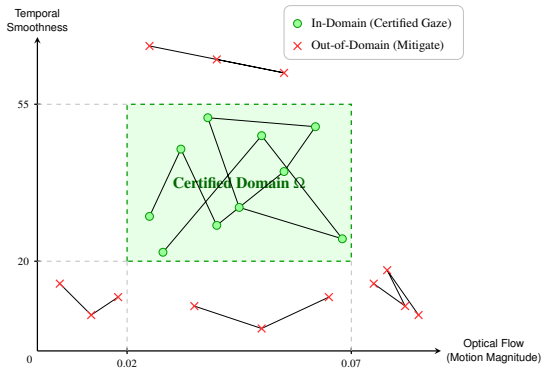

Figure 7: Visualizing the Domain-Conditioned Safe Operating Range. Based on exact binomial confidence bounds, the runtime reliability layer defines a bounded region Ω over observable scene statistics. When telemetry windows (green dots) fall within limits such as optical flow $\in [0.02, 0.07)$ and temporal smoothness $\in [20, 55]$, sustained OOI attention is certified with $p \geq 0.82$. Perturbations that push the system state outside these bounds (red crosses) trigger immediate mitigation to prevent gaze degradation.

This domain captures scenes with moderate motion and sufficient temporal continuity, as virsualized by Fig. 7. The result acts as a conditional assurance on attention-relevant behavior. When the rendering pipeline stays in this stable regime, the system can trust gaze-driven OOI response substantially more than it could from average RMSE metrics alone. Consequently, MR platforms can deploy this certified domain as a real-time admission control mechanism for attention-aware features.

7.4 Bounded Divergence Under Perturbations

The second guarantee quantifies robustness to latency-based cognitive perturbation. Table 3 reports exact lower confidence bounds for bounded divergence over 1.0 second windows across three FPS bins. Each bin relies on a minimal operating domain defined over observable scene statistics.

The sensitivity analysis in Table 4 shows that the thresholding policy is stable across temporal aggregation choices. In the low and medium bins, all three window lengths retain perfect empirical success and only the conservative lower bound decreases slightly because longer windows yield fewer trials. In the strong-attack bin, the empirical success rate remains tightly clustered between 0.728 and 0.731, with the lower bound varying by less than two percentage points. This indicates that the certificate is not brittle to the exact choice of T. Its dominant sensitivity comes from attack severity rather than from modest changes in window duration.

Several structural patterns matter for system deployment. First, the guarantee naturally weakens as attack severity increases, which demonstrates credible and expected system behavior. Second, even in the highest severity bin, the certified lower bound remains exceptionally strong provided the system state remains inside the vetted domain. Third, the constrained statistics are system-meaningful. Brightness flicker, spectral entropy, spatial frequency, and edge intensity are all parameters that the rendering pipeline or UI overlay manager can directly monitor and actively influence. This directly transitions the guarantee from a descriptive metric into an operational reliability layer.

8 Discussion

The central systems insight of this work is that gaze reliability must be reframed as a dynamic admission control problem. Rather than relying on static, global error metrics that inevitably fail under adversarial conditions, a deployed MR system must continuously evaluate whether its current operating state belongs to a certified regime. This paradigm shift fundamentally alters how attention-aware adaptation should be implemented. By tying confidence directly to measurable scene statistics and telemetry gaps, the framework ensures that task-critical decisions are grounded in verifiable human behavior.

When the runtime state resides within the certified domain, the system can confidently execute default gaze-driven interactions. Conversely, when service disruptions push the system out-of-domain, the architecture provides a deterministic trigger for active mitigation. Rather than allowing interaction quality to silently degrade, the MR system can dynamically adapt its rendering pipeline to re-enter the safe envelope. Actionable responses include reducing rendering complexity to recover temporal smoothness, simplifying UI overlays to lower brightness flicker, enlarging interface targets, or temporarily falling back to multimodal interaction. Under severe latency attacks, this domain-conditioned approach functions as a robust defense layer, transforming unpredictable system failures into graceful degradation.

The ultimate implication of this framework is a highly deployable architectural pattern for mission-critical XR. By coupling time-aware gaze forecasting with exact statistical certificates, developers are equipped to build systems that autonomously decide exactly when to trust gaze inference and when to enforce conservative safety policies.

9 Conclusion

This paper unifies predictive gaze modeling, mission-level assurance, and system robustness into a single cohesive mixed-reality architecture. We first empirically established the critical link between rendering disruptions and the structural breakdown of human visual attention. To protect against these disruptions, we introduced a time-aware forecasting model that explicitly mitigates the impact of irregular telemetry and latency perturbations. Building upon this robust baseline, we developed a domain-conditioned certification framework. This framework successfully translates continuous gaze predictions into strict, coverage-qualified probabilistic guarantees for timely target orientation and sustained fixation. By defining measurable safe operating domains over scene statistics, our approach ensures that mixed-reality systems can actively monitor, verify, and sustain attention-aware interactions even in highly contested environments.

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Table 3: Certified divergence guarantees at $T = 1.0$ s and confidence 0.9.

Bin	Domain constraints	Coverage	$p_{\mathrm{LB}}(\beta{=}0.20)$	$p_{\mathrm{LB}}(\beta{=}0.25)$	$p_{\mathrm{LB}}(\beta{=}0.30)$
Low (52.5 to 59.9 FPS)	brightness_flicker $\in [0.209, 0.963]$	0.802	0.998	0.998	0.998
Mid (45.0 to 52.5 FPS)	spectral_entropy $\in [6.822, 7.308]$, spatial_frequency $\in [312.2, 980.25]$	0.755	0.981	0.994	0.998
High (30.0 to 45.0 FPS)	edge_intensity $\in [4.239, 7.812]$, spatial_frequency $\in [312.2, 980.25]$	0.686	0.945	0.970	0.991

Table 4: Window-length sensitivity for the bounded-divergence certificate at $\beta = 0.25$ and confidence 0.9.

Attack bin	T (s)	N	K	$\hat{p}$	p_{LB}
Low	0.5	952	952	1.000	0.998
Low	1.0	475	475	1.000	0.995
Low	1.5	316	316	1.000	0.993
Medium	0.5	952	952	1.000	0.998
Medium	1.0	475	475	1.000	0.995
Medium	1.5	316	316	1.000	0.993
High	0.5	952	694	0.729	0.710
High	1.0	475	347	0.731	0.703
High	1.5	316	230	0.728	0.693

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